Project Documentation

GLOBAL FINANCIAL MARKET INTELLIGENCE HUB

An Azure Data Engineering Platform for Cross-Asset Financial Intelligence

Platform Used:

Azure Data Factory · ADLS Gen2 · Databricks & Delta Lake · Azure SQL · Power BI

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1. Executive Summary

The Global Financial Market Intelligence Hub is an Azure data engineering platform that brings equities, ETFs, and funds from the FinanceDatabase repository into one common, taxonomy-mapped, quality-scored analytics layer — served through Azure SQL and a five-page Power BI reporting hub. The final catalog covers 66,000+ instruments across 10 exchanges, 95 countries, 18 currencies, and 9 normalized financial themes, every one of them traceable back to its original source classification.

2. Problem

Financial research teams often work with data from multiple asset classes, markets, exchanges, and geographic regions. The trouble is that classification structures are rarely consistent across those asset classes — an equity is typically classified by Sector → Industry Group → Industry, while an ETF or fund is classified by Family → Category Group → Category. Because these hierarchies don't line up, analysts can't easily answer cross-asset questions such as “how broadly is the Technology theme represented across equities, ETFs, and funds?” or “which countries have the strongest market representation for Healthcare?”

What was needed was a centralized platform that could integrate, standardize, and classify financial instrument data across asset classes — without losing the original source information behind each classification.

3. Solution & Objectives

The platform ingests raw financial datasets, retains immutable source snapshots, standardizes heterogeneous schemas into one common instrument model, and maps every source classification onto a shared, organization-defined Common Financial Taxonomy — while preserving the original classification alongside the normalized one. Every mapping carries a confidence score, every record carries a data-quality assessment, and every result is served through both a relational warehouse and an interactive Power BI hub.

Scope

The platform covers three asset classes — equities, ETFs, and funds — across 10 exchanges: BER, BSE, FRA, GER, JPX, LSE, NSE, SHZ, VIE, and NYQ, with coverage varying by exchange (for example, NSE carries equities only).

Key objectives

- Ingest financial datasets from the FinanceDatabase GitHub repository into immutable, snapshot-partitioned raw storage.
- Standardize heterogeneous asset-class schemas into a common financial instrument model.
- Standardize reference data — countries, currencies, and exchanges — into clean dimensions.

- Build a Common Financial Taxonomy and map every source classification onto it, preserving the original classification.
- Score every taxonomy mapping for confidence, and every dataset for completeness, validity, and quality.
- Isolate invalid or unmapped records into a quarantine layer rather than discarding them.
- Produce reusable, business-ready Gold data products with Delta Lake versioning and time travel.
- Serve the curated model through Azure SQL and an interactive Power BI Market Intelligence Hub.

4. Architecture Overview

The platform follows a medallion architecture end to end: Azure Data Factory ingests raw source files into a Bronze layer on ADLS Gen2; Databricks notebooks transform Bronze into standardized, taxonomy-mapped, quality-scored Silver and Gold Delta tables; Azure SQL loads the Gold layer into a star schema; and Power BI serves the final analytics experience. Every Azure resource for the project lives in a single resource group.

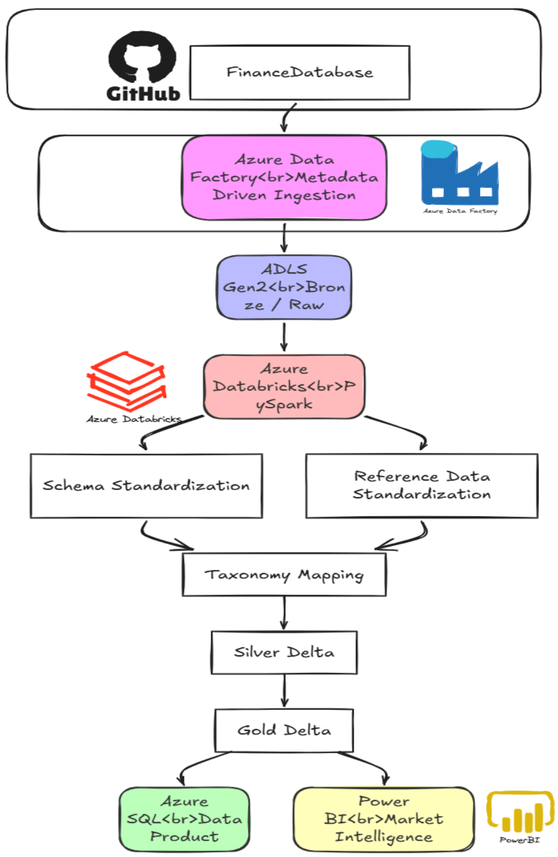


Figure 0 — End-to-end platform data flow, from raw source to Power BI.

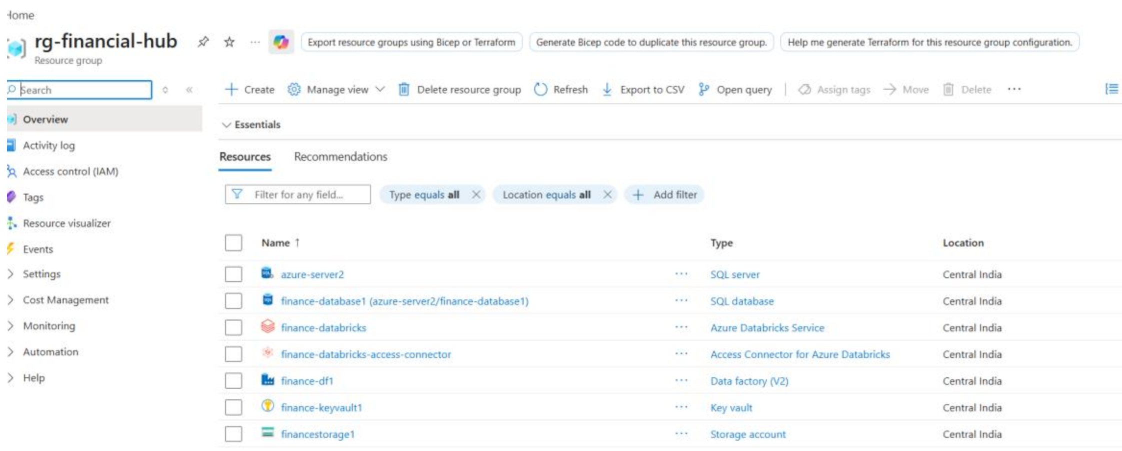


Figure 1 — All Azure resources for the platform: Data Factory, Databricks workspace and access connector, SQL server and database, Key Vault, and the ADLS Gen2 storage account, grouped under rg-financial-hub.

5. Technology Stack

Technology	Role in the Platform
Azure Data Factory	Metadata-driven orchestration for ingestion (source → Bronze) and for loading Gold/Silver into Azure SQL
Azure Data Lake Storage Gen2	Bronze / Silver / Gold / Quarantine medallion storage
Azure Databricks (PySpark)	Schema standardization, reference-data standardization, taxonomy mapping, data quality, Gold processing
Delta Lake	ACID transactions, schema enforcement, MERGE/upserts, versioning, and time travel on Silver and Gold
Azure SQL Database	Curated star-schema serving layer (staging + dimensional model) for Power BI
Power BI	Five-page Global Market Intelligence Hub for interactive analysis
Managed Identity / Microsoft Entra ID	End-to-end authentication across ADF, Databricks, and Azure SQL — no stored credentials
Azure Key Vault	Secret storage for service principal credentials used in Databricks-to-SQL connectivity

6. Data Lake — Medallion Architecture

The data lake is organized into four containers that mirror the medallion pattern: Bronze for raw, immutable source snapshots; Silver for standardized, cleaned Delta tables; Gold for business-ready analytical datasets; and Quarantine for records that fail validation or classification, kept for review rather than discarded.

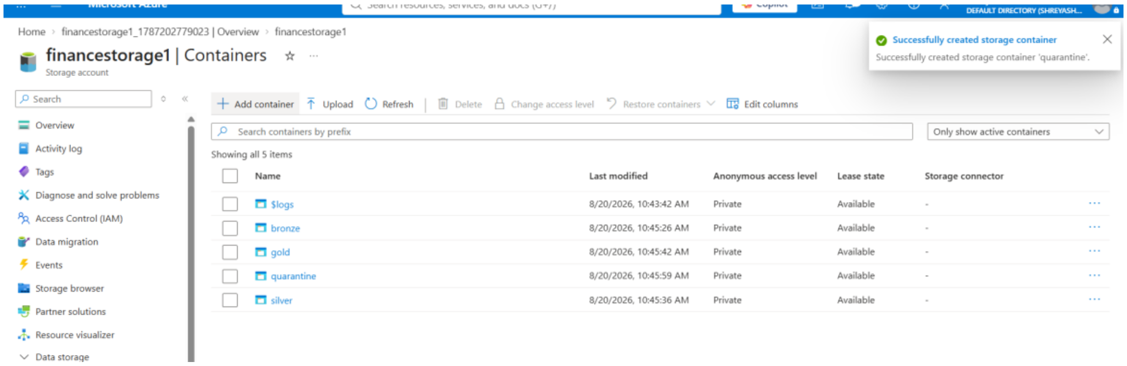


Figure 2 — The four containers in the ADLS Gen2 storage account: bronze, silver, gold, and quarantine.

Bronze — immutable source snapshots

Every ingestion run lands data into a date-partitioned snapshot folder rather than overwriting the previous run, which keeps the full ingestion history reproducible. Below is the Bronze folder for equities on the BER exchange, showing five days of daily snapshots.

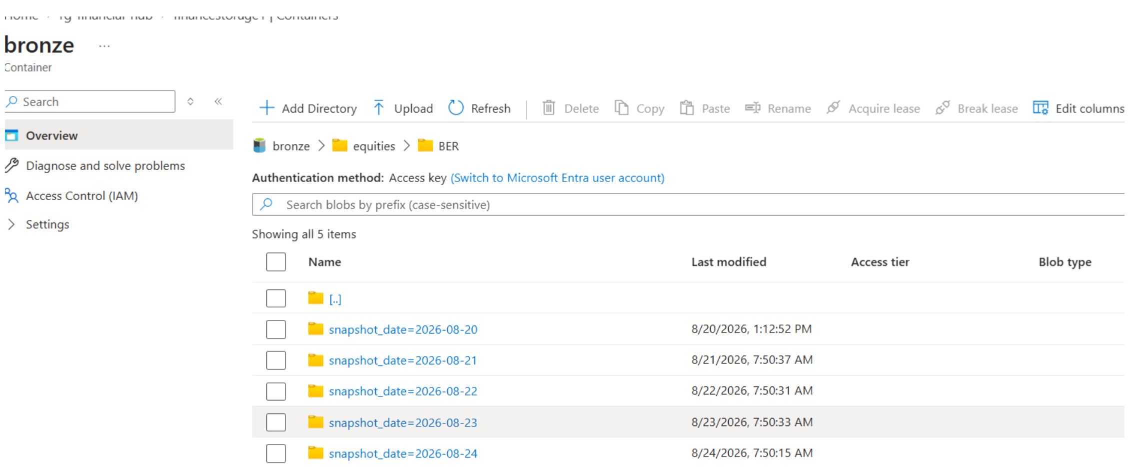


Figure 3 — Bronze snapshot partitions for bronze/equities/BER, one folder per ingestion date.

Gold — business-ready data products

The Gold layer holds five curated, analytics-ready Delta tables built from Silver: the financial instrument catalog, market universe summary, theme universe, cross-asset theme comparison, and the data-quality run history.

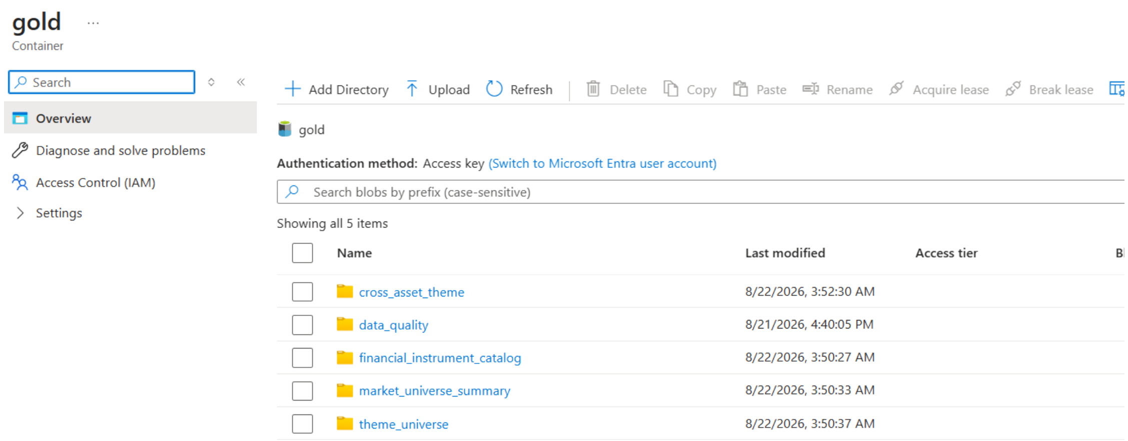


Figure 4 — The five Gold Delta tables: cross_asset_theme, data_quality, financial_instrument_catalog, market_universe_summary, and theme_universe.

Quarantine — isolating rather than discarding

Records that fail validation or cannot be mapped to the taxonomy are not dropped — they are written to a quarantine container split into invalid_records and unmapped_classifications, so they remain available for review and reprocessing.

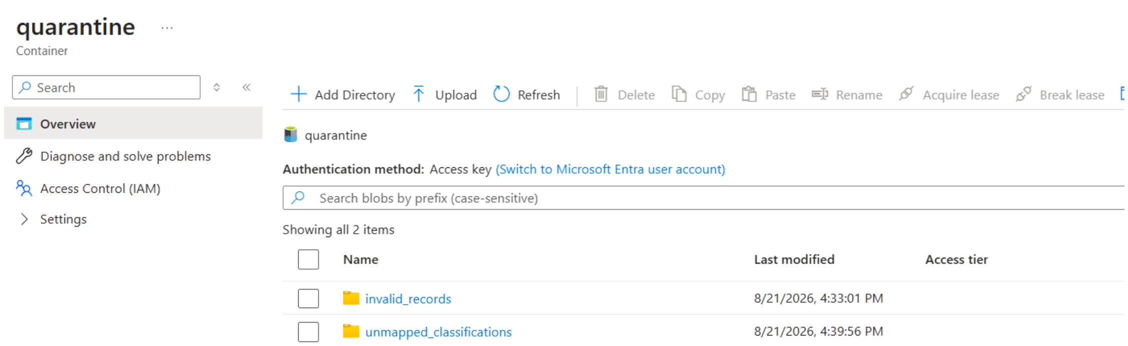


Figure 5 — The quarantine container, holding invalid_records and unmapped_classifications separately.

7. Ingestion Layer — Azure Data Factory

Ingestion is fully metadata-driven: a control table lists every source dataset, its format, and its target Bronze path, and the pipeline dynamically loops over active rows rather than hard-coding one activity per dataset. Authentication throughout the factory uses Managed Identity rather than stored secrets.

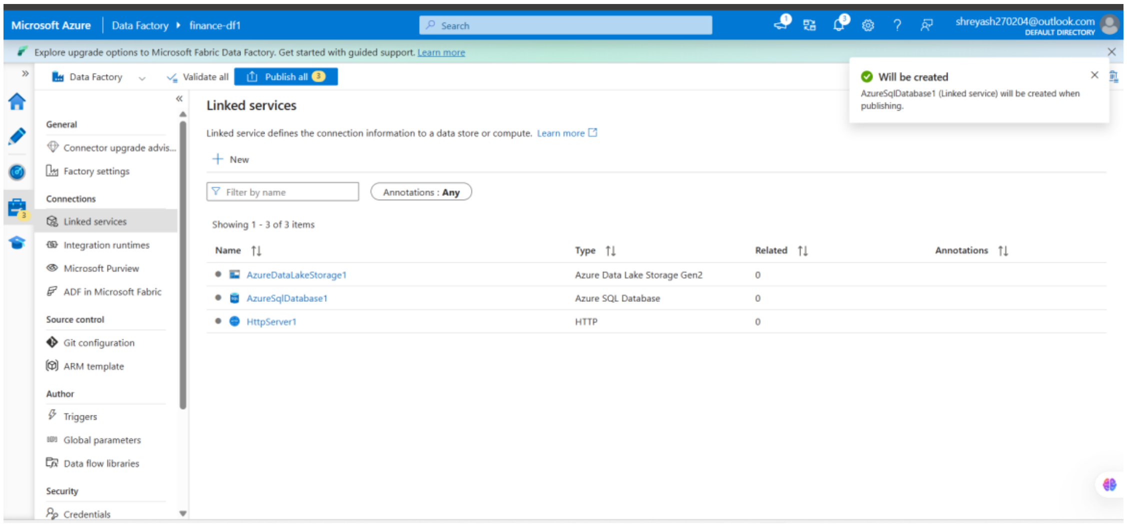


Figure 6 — ADF linked services: ADLS Gen2, Azure SQL Database, and the HTTP source for FinanceDatabase — no credentials stored on any of them.

Bronze ingestion pipeline

PL_Bronze_Ingestion reads the ingestion control table with a Lookup activity, then loops through each active row with a ForEach activity, copying each source dataset into its Bronze target path.

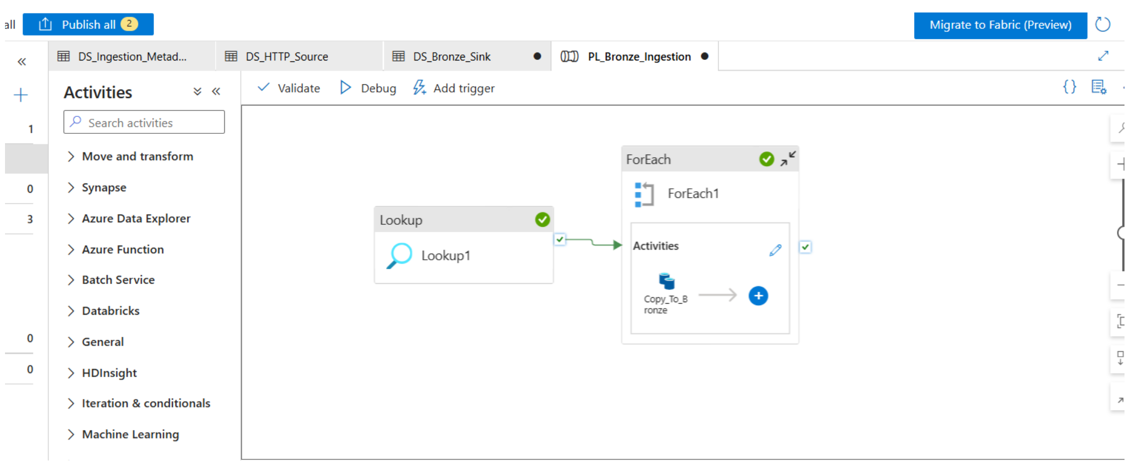


Figure 7 — PL_Bronze_Ingestion: Lookup reads the control table, then ForEach copies each dataset to Bronze.

Every iteration of the pipeline is logged and auditable. A full debug run across all configured datasets completes with every Copy activity reporting Succeeded, each in 15–20 seconds.

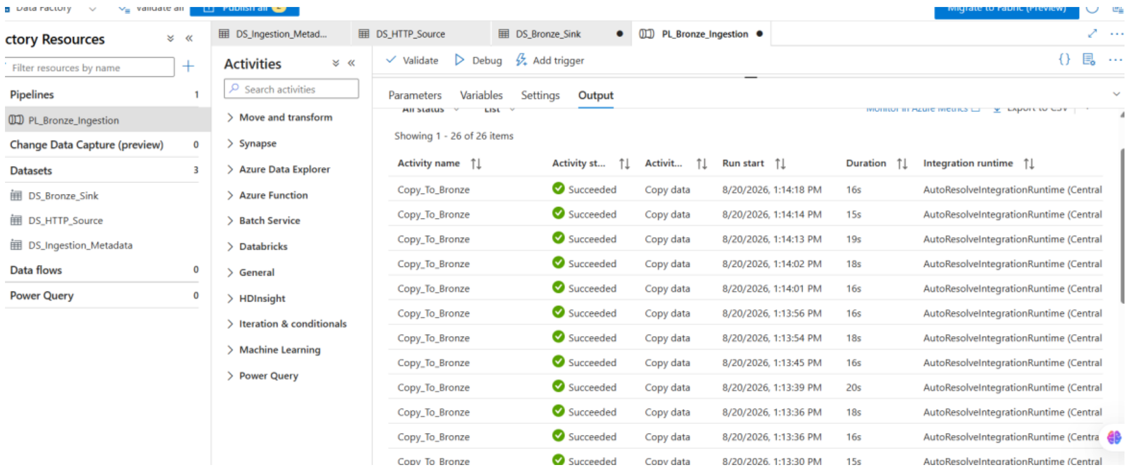


Figure 8 — Pipeline execution output: 26 Copy_To_Bronze iterations, all succeeded.

Gold-to-SQL load pipeline

A second pipeline, PL_Gold_To_SQL, loads the curated Gold and Silver datasets into the Azure SQL staging schema and then triggers the stored procedure that builds the star schema — all without any Databricks compute in the loop (see the cost strategy in Section 11).

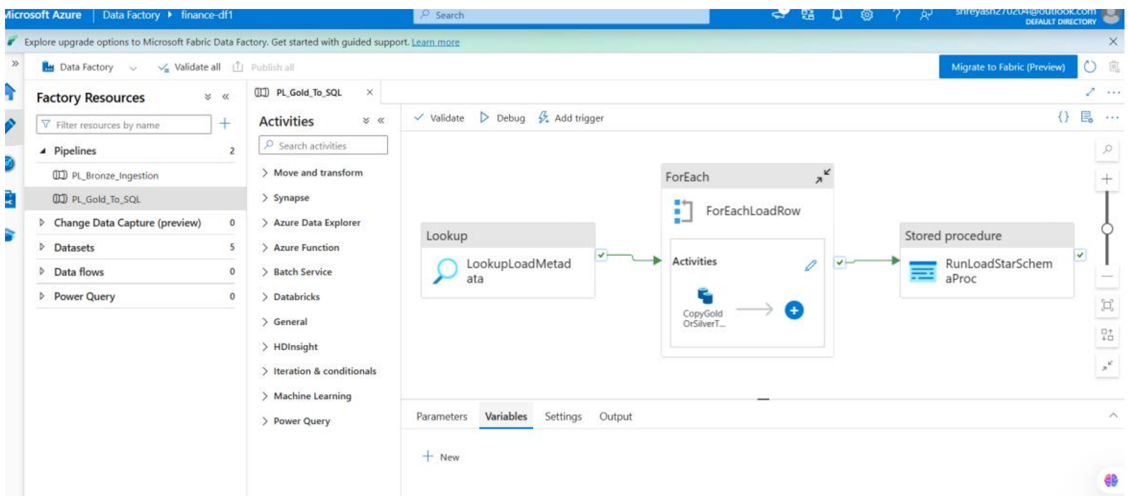


Figure 9 — PL_Gold_To_SQL: Lookup reads the load metadata, ForEach copies each table into staging, then a Stored Procedure activity builds the star schema.

8. Transformation Layer — Azure Databricks

All transformation logic runs as seven config-driven PySpark notebooks in Databricks, moving data from Bronze through Silver to Gold. Keeping the mapping and configuration values in a shared utilities module (rather than hard-coded throughout the notebooks) keeps the taxonomy and reference-data logic reusable and auditable.

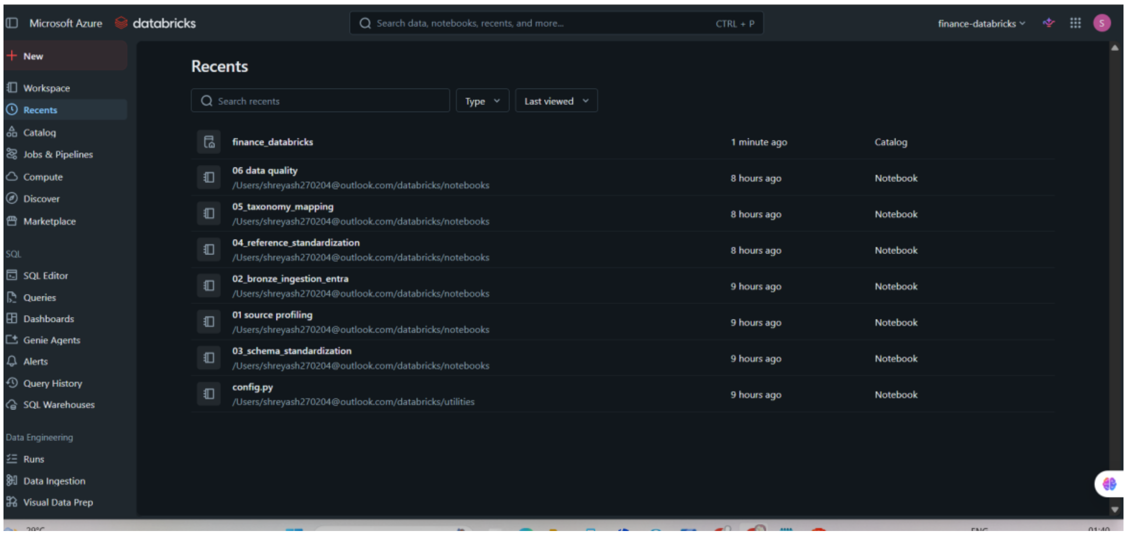


Figure 10 — The Databricks workspace notebooks: source profiling, bronze auditing, schema standardization, reference-data standardization, taxonomy mapping, data quality, and Gold processing.

Notebook pipeline

Notebook	Purpose
01 — Source Profiling	Profiles columns, nulls, duplicates, and classification structures directly against raw Bronze data
02 — Bronze Ingestion Audit	Cross-checks ADF's ingestion control table against what actually landed in Bronze
03 — Schema Standardization	Builds the common silver_financial_instrument model plus asset-specific extension tables
04 — Reference Standardization	Builds dim_exchange, dim_country, and dim_currency
05 — Taxonomy Mapping	Maps source classifications (sector, category, etc.) to the Common Financial Taxonomy with confidence scoring
06 — Data Quality	Runs completeness, validity, duplicate, and taxonomy checks; routes failures to quarantine
07 — Gold Processing	Builds the four Gold tables, computes the Theme Coverage Score, and demonstrates Delta time travel

Silver layer — standardized instrument model

The common silver_financial_instrument table unifies equities, ETFs, and funds into one instrument model while asset-specific attributes (sector/industry for equities, category/family for ETFs and funds) remain in separate extension tables — preserving source fidelity while still enabling cross-asset analysis.

9. Common Financial Taxonomy & Data Quality

The central engineering challenge of the platform is reconciling incompatible classification systems across asset classes into one shared taxonomy — nine normalized financial themes: Technology, Healthcare, Energy, Financial Services, Industrials, Consumer, Real Estate, Utilities, and Materials — without losing the original source classification. Every instrument keeps both its source_classification and its normalized_theme, along with a mapping_method and mapping_confidence score, so a normalized theme is always traceable back to the raw sector, category, or family value it came from. Mapping is config-driven, and field priority is chosen per asset type based on which source field empirically yields better theme coverage — equities prefer industry_group over sector, while ETFs and funds prefer category_group over category.

Every pipeline run scores each dataset for completeness, validity against reference dimensions, taxonomy quality, and duplication — feeding a data-quality history table that tracks trends run over run.

Dataset	Total Records	Valid Records	Quality Score
Equities	40,549	40,174	99.08%
ETFs	13,782	13,782	100.00%
Funds	11,774	11,770	99.97%

Each theme also carries a weighted Theme Coverage Score — combining asset-class breadth, country reach, exchange reach, and instrument scale — surfaced directly on the Cross-Asset Explorer page of the Power BI report, where Financial Services currently leads the ranking with the broadest theme coverage.

10. Engineering Challenges & Solutions

Three issues surfaced during development stood out as genuine engineering problems, each spanning a different layer of the platform — ingestion, transformation, and serving.

Blob-URL Ingestion Bug — Ingestion Layer

Bronze ingestion pointed at GitHub's HTML "blob" URLs instead of the raw file endpoint, so every file that landed in Bronze was actually a webpage, not a CSV — and the pipeline still reported success, since the corruption was silent. Fixed by repointing the source to raw.githubusercontent.com and re-ingesting every affected dataset. This is why direct data verification, not just a green pipeline status, is treated as a required step.

Multi-Line CSV Parsing — Transformation Layer

Company description fields in the source data contain embedded line breaks inside quoted CSV values, which Spark's CSV reader doesn't handle by default — silently shifting every column to the right for affected rows, across the whole dataset. Caught when a discovery query showed currency codes and sentence fragments inside the sector column. Fixed with multiLine and escape options on every CSV read.

Managed Identity Permission Gaps — Serving Layer

Azure SQL's deliberately vague "object does not exist or you do not have permission" error masked two separate causes at once: a missing ALTER grant on the staging schema (needed for TRUNCATE TABLE), and a server-level "Entra ID only" setting silently blocking the dedicated Power BI login. Each was diagnosed and fixed individually — targeted GRANT statements and a server-setting change — rather than broadening access permissions as a blanket fix.

11. Serving Layer — Azure SQL

The Gold and Silver Delta tables are loaded into a star-schema Azure SQL database, giving Power BI a fast, relational model to query instead of reading Delta files directly. Two schemas separate the landing zone from the curated model: stg holds a truncate-and-reload staging copy of each source table, and dbo holds the curated dimensions, facts, and business views that Power BI actually connects to.

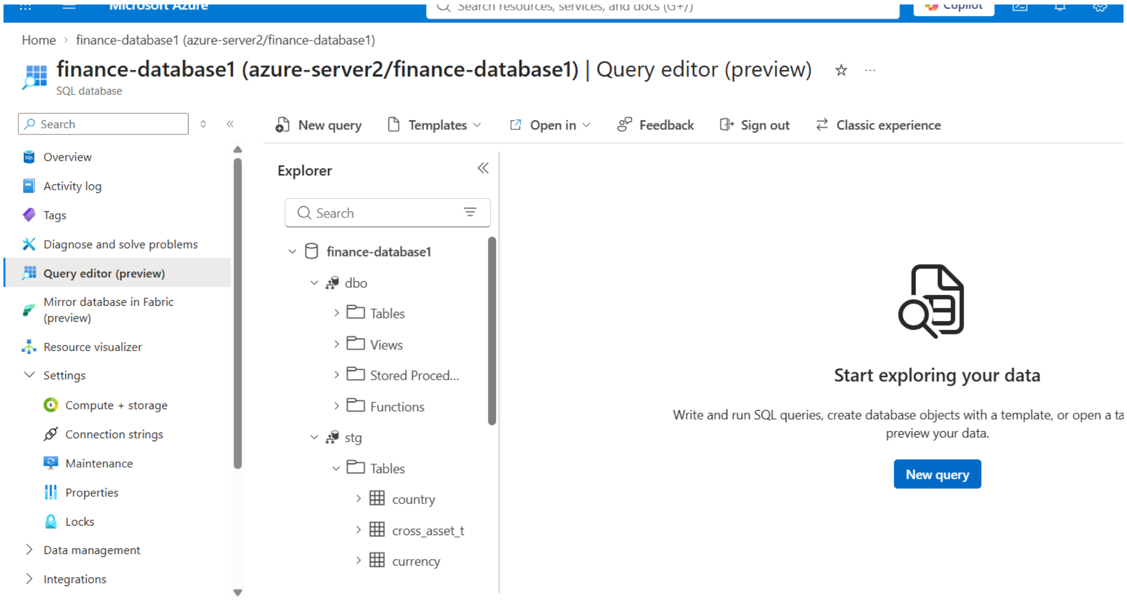


Figure 11 — The Azure SQL database in the Query editor, showing the dbo (curated) and stg (staging) schemas.

Cost-conscious design

Rather than routing the Gold-to-SQL load through another Databricks job, the ADF Copy activity reads the Parquet files underlying the Gold and Silver Delta tables directly off ADLS Gen2, using ADF's own integration runtime — billed per copy-minute rather than cluster time. Setting recursive: false on the file read deliberately skips Delta's _delta_log folder, so no Delta-aware connector or extra Databricks linked service was needed at all. All natural-key-to-surrogate-key resolution runs as T-SQL stored procedures inside Azure SQL — compute that was already being paid for as part of the database, adding no incremental cost.

Star schema

Table	Rows	Notes
FactInstrument	66,105	Grain: one row per instrument, sourced from the Gold financial instrument catalog
FactDataQualityRun	3	One row per dataset (equities/ETFs/funds) per pipeline run
DimDate	~2,500	Generated locally in T-SQL
DimCountry	95	Deduplicated by country name
DimExchange	10	Deduplicated by exchange code

Table	Rows	Notes
DimCurrency	18	Verified directly against the source — the complete set actually used across all instruments
DimAssetType	3	Equity / ETF / Fund
DimTheme	9	Matches the Common Financial Taxonomy exactly

18 curated business views (dbo.vw_*) sit on top of the dimensional model, pre-built to answer the platform's core business questions across global market, cross-asset, taxonomy, and data-quality themes — these views, not the raw dimension and fact tables, are what Power BI imports.

12. Reporting Layer — Power BI Market Intelligence Hub

The Market Intelligence Hub is a five-page Power BI report, connected to Azure SQL through a dedicated read-only login kept separate from the write-capable Managed Identity used by the load pipeline. All five pages are navigable from a single published report.

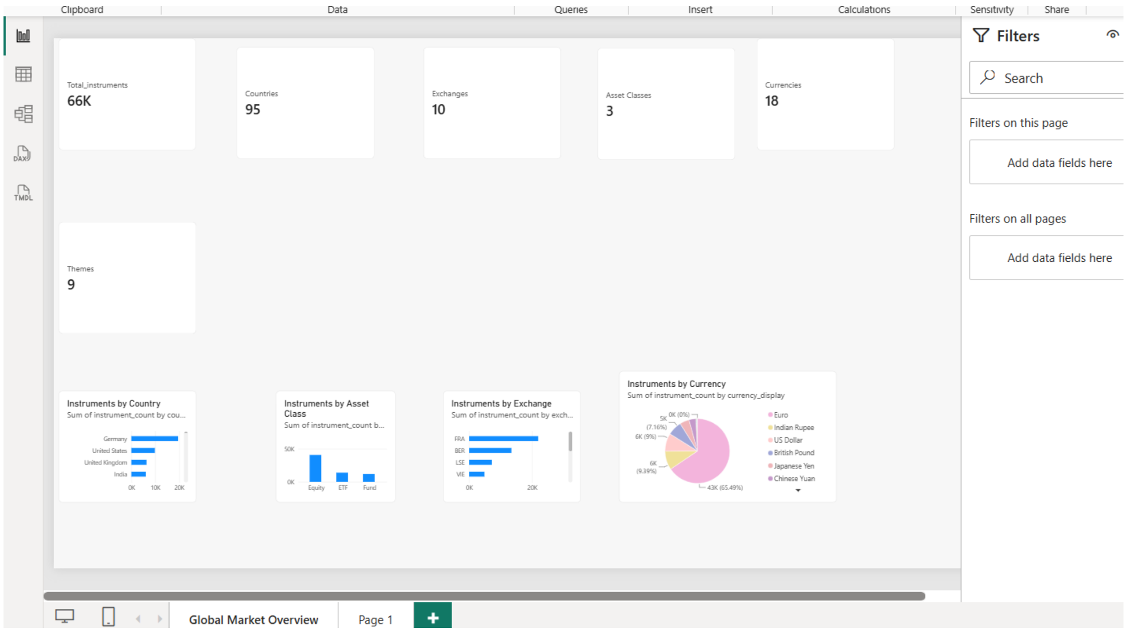


Figure 12 — The published report, showing all five page tabs: Global Market Overview, Theme Intelligence, Cross-Asset Explorer, Geographic Intelligence, and Data Product Health.

Page 1 — Global Market Overview

Six headline KPIs — Total Instruments, Countries, Exchanges, Asset Classes, Currencies, and Themes — sit alongside breakdowns of instrument volume by country, exchange, asset class, and currency, giving a single top-level view of the entire financial universe the platform covers.

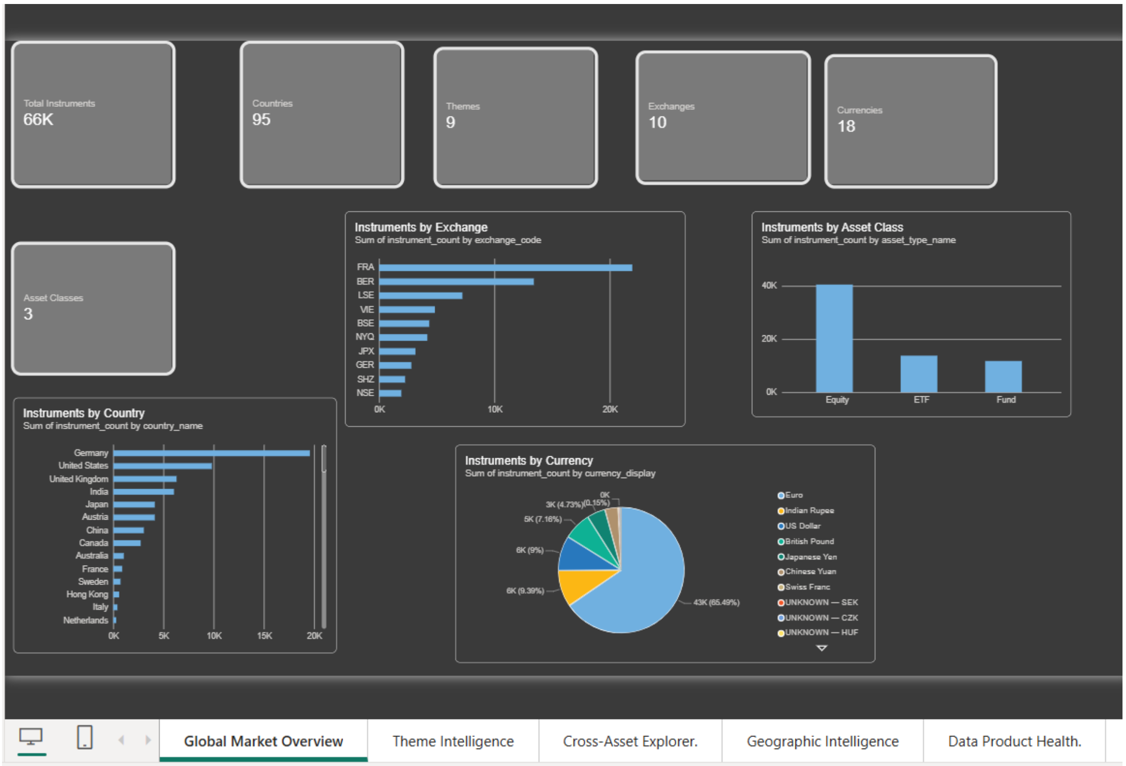


Figure 13 — Global Market Overview, authored in Power BI Desktop: 66K total instruments across 95 countries, 10 exchanges, 3 asset classes, and 18 currencies.

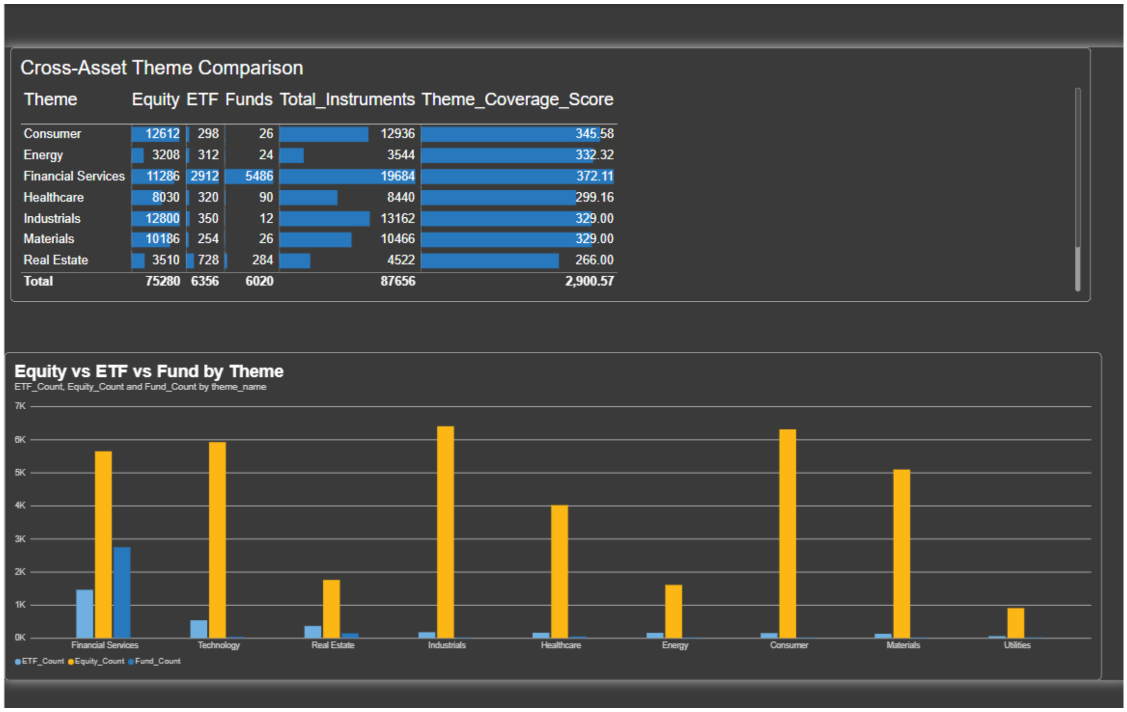
Page 2 — Theme Intelligence

Selecting a financial theme filters the report down to that theme's footprint — instruments in theme, countries covered, exchanges covered, and its Theme Coverage Score, broken down by country, asset class, and exchange.

Figure 14 — Theme Intelligence, with theme-level KPIs and breakdowns by country, asset class, and exchange.

Page 3 — Cross-Asset Explorer

This page directly answers the platform's founding question: how does the same financial theme compare across equities, ETFs, and funds? A side-by-side comparison table and grouped chart rank every theme by total instrument count and Theme Coverage Score.


Cross-Asset Theme Comparison

Theme	Equity	ETF	Funds	Total_Instruments	Theme_Coverage_Score
Consumer	12612	298	26	12936	345.58
Energy	3208	312	24	3544	332.32
Financial Services	11286	2912	5486	19684	372.11
Healthcare	8030	320	90	8440	299.16
Industrials	12800	350	12	13162	329.00
Materials	10186	254	26	10466	329.00
Real Estate	3510	728	284	4522	266.00
Total	75280	6356	6020	87656	2,900.57

Figure 15 — Cross-Asset Explorer: Financial Services leads with the broadest theme coverage, followed by Consumer and Technology.

Page 4 — Geographic Intelligence

Country and theme slicers drive a shared set of visuals — instruments by asset class, by exchange, and by region — letting a researcher explore the financial universe from a geographic starting point instead of a thematic one.

Figure 16 — Geographic Intelligence: country and theme slicers driving instrument breakdowns by asset class, exchange, and region.

Page 5 — Data Product Health

The final page turns the platform's own data-quality process into a dashboard: overall mapped percentage, total instruments, unmapped classifications, low-confidence mappings, duplicate and invalid record counts, and a per-dataset pipeline status table with the latest quality scores.

Figure 17 — Data Product Health: live quality KPIs and the latest pipeline run status for equities, ETFs, and funds.

13. Security

No credentials are hard-coded anywhere in the platform. Authentication is handled end to end through:

- Managed Identity for ADF's access to ADLS Gen2 and Azure SQL, granted least-privilege RBAC roles (e.g. Storage Blob Data Contributor).
- A Databricks system-assigned Managed Identity, via an Access Connector, for ADLS access from PySpark notebooks.
- Microsoft Entra ID Service Principal authentication (JDBC ActiveDirectoryServicePrincipal) for the Databricks-to-SQL control-table audit, with credentials held in Key Vault through a Databricks secret scope.
- A dedicated, read-only SQL login for Power BI, kept intentionally separate from ADF's write-capable Managed Identity as a clean security boundary between the load path and the reporting path.

14. Business Questions Answered

The platform's 18 curated Azure SQL views and five Power BI pages together answer questions across four areas:

Category	Example Questions
Global Market	Which countries and exchanges have the largest instrument universe? Which currencies and asset classes dominate?
Cross-Asset	Which themes are represented across the most asset classes and the broadest geography?
Taxonomy	What percentage of records map to the common taxonomy, and which classifications remain unmapped or low-confidence?
Data Quality	Which dataset has the highest quality-issue rate, and how has quality changed between pipeline runs?

15. Learnings

Building this platform end to end surfaced practical lessons across every layer of the stack:

- Designing a metadata-driven ADF pipeline that scales to new datasets without writing new pipeline logic.
- Building a full medallion architecture — Bronze, Silver, Gold, and Quarantine — on Delta Lake with real versioning and time travel.
- Reconciling inconsistent classification systems from different asset classes into one shared taxonomy without losing the original source meaning.
- Debugging real Azure issues — Managed Identity permission errors, silent CSV corruption — by verifying data directly rather than trusting a green pipeline status.
- Designing for cost as deliberately as for correctness — reading Parquet directly off ADLS instead of adding an unnecessary Databricks compute step.
- Building a star schema and a layer of business views that make a Power BI report fast and simple to author against.
- Turning a technical data-quality process into an executive-facing Power BI page that end users actually read and trust.

16. Conclusion

The Global Financial Market Intelligence Hub takes heterogeneous financial datasets — each with its own schema and classification system — and turns them into one trusted, versioned, reusable financial intelligence layer, without losing the meaning of the original source data. Every stage of the pipeline, from raw ingestion through taxonomy mapping to the final Power BI dashboards, is traceable, auditable, and built on credential-free, Managed Identity–based authentication throughout.

Beyond the pipeline itself, the project demonstrates the discipline that real data engineering requires: verifying data directly rather than trusting green checkmarks, isolating rather than discarding bad records, and designing for cost as deliberately as for correctness.